ResuMate: AI-Powered Career Intelligence Platform

Overview

ResuMate is an AI-powered Career Intelligence Platform designed to go beyond traditional Applicant Tracking Systems (ATS).

Most ATS platforms rely heavily on keyword matching and often fail to accurately process scanned resumes, image-based CVs, or unconventional resume formats. Even when information is extracted successfully, these systems typically assume that the skills listed on a resume accurately reflect a candidate's capabilities.

ResuMate addresses these challenges by combining OCR, NLP, Machine Learning, Assessment Generation, Recommendation Systems, and Predictive Analytics into a unified career intelligence ecosystem.

The platform is designed to:

- Extract information from any resume format
- Identify candidate skills and experience
- Verify skills through AI-generated assessments
- Predict suitable job roles
- Estimate salary ranges
- Generate personalized learning roadmaps
- Provide long-term career guidance

Team Structure

Shivansh Bhadauriya (AI / ML Engineer)

Responsible for:

- OCR Pipeline
- Resume Parsing
- NLP Extraction
- Skill Identification
- Assessment Generation
- Candidate Intelligence Engine
- Job Prediction
- Salary Prediction
- User Clustering
- Roadmap Generation
- FastAPI ML Microservice

Tejas (Full Stack Engineer)

Responsible for:

- React Frontend
- Node.js Backend
- Authentication

- API Gateway
- MongoDB Design
- Dashboard Development
- State Management
- API Integration

High-Level Architecture

```
User Uploads Resume
        │
        ▼
React Frontend
        │
        ▼
Node.js Backend
        │
        ▼
MongoDB
        │
        ▼
FastAPI ML Service
        │
        ▼
Resume Intelligence Pipeline
        │
        ▼
Recommendations & Insights
```

Resume Intelligence Pipeline

The AI layer consists of multiple stages that progressively transform an uploaded document into actionable career intelligence.

```
Resume
   │
   ▼
Document Ingestion
```

```
    |
    ▼
OCR & Text Extraction
    |
    ▼
NLP Information Extraction
    |
    ▼
Candidate Profile Generation
    |
    ▼
Assessment Generation
    |
    ▼
Candidate Evaluation
    |
    ▼
Recommendation Engine
```

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1. Document Ingestion Layer

The first responsibility of the system is handling multiple resume formats.

Supported formats:

- PDF
- Scanned PDF
- DOCX
- PNG
- JPG
- JPEG

A smart dispatcher determines the document type and routes it through the appropriate extraction pipeline.

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2. OCR & Text Extraction

Many resumes arrive as scanned documents or images.

Traditional parsers fail in these situations.

To solve this problem, ResuMate uses an OCR pipeline.

Technologies

- pdfplumber
- pdf2image
- Tesseract OCR

• Pillow

Workflow

```
Resume
    │
    ▼
Document Type Detection
    │
 ┌──┴──┐
 │     │
PDF   Image
 │     │
 ▼     ▼
Parser OCR
 │
 ▼
Raw Text
```

OCR Optimization

Before OCR processing:

- Grayscale Conversion
- Lanczos Upscaling
- Contrast Enhancement are applied to improve extraction accuracy.

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3. NLP Information Extraction

Once text is available, the system converts unstructured content into structured information.

The extraction layer combines Regex and NLP techniques.

Regex Extraction

Identifies:

• Email Addresses
• Phone Numbers
• LinkedIn Profiles
• GitHub Profiles

NLP Extraction

Identifies:

• Technical Skills
• Soft Skills
• Frameworks
• Tools

- Education
- Projects
- Experience

Output

```
{
  "email": "...",
  "phone": "...",
  "skills": [...],
  "education": [...],
  "projects": [...]
}
```

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4. Candidate Profile Generation

The extracted information is transformed into a structured candidate profile.

This profile acts as the central representation of a user within the platform.

The profile contains:

- Contact Information
- Skills
- Projects
- Experience
- Education
- Resume Metadata

This profile becomes the foundation for all future recommendations.

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5. Assessment Generation Engine

Traditional ATS systems trust whatever appears on a resume.

ResuMate attempts to verify candidate skills.

Workflow

```
Extracted Skills
        |
        ▼
Prompt Builder
        |
        ▼
LLM
        |
        ▼
Assessment Questions
```

Example:

Skills:

- Python
- Machine Learning
- FastAPI

Generated Output:

- Python Questions
- ML Questions
- Backend Questions

Each candidate receives a personalized assessment based on their profile.

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6. Candidate Evaluation Layer

After the assessment is completed, responses are evaluated.

The system generates:

- Skill Scores
- Competency Scores
- Overall Candidate Score Example:

```
Python            85%

Machine Learning  78%

FastAPI           82%



Overall Score     81%
```

The objective is to create a verified candidate profile rather than relying solely on self-reported skills.

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7. Job Prediction Engine

The job prediction module identifies career paths most aligned with the candidate's verified skills.

Inputs

- Skills
- Projects
- Education
- Assessment Results
- Candidate Score

Output

```
Machine Learning Engineer

AI Research Intern

Data Analyst

Backend Developer
```

Future versions will use embedding-based semantic matching to compare candidate profiles with industry job descriptions.

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8. Salary Prediction Engine

The salary prediction module estimates realistic compensation ranges.

Inputs

- Skills
- Experience
- Education
- Location
- Assessment Performance

Potential Models

- Random Forest Regressor
- XGBoost
- Gradient Boosting

Output

```
Estimated Salary Range

₹8-12 LPA
```

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9. User Clustering Engine

Candidates with similar characteristics are grouped into clusters.

Purpose

- Better Recommendations
- Peer Benchmarking
- Personalized Roadmaps
- Career Insights

Possible Algorithms

- K-Means
- DBSCAN
- Hierarchical Clustering Example Clusters:

- AI Engineers
- Data Scientists
- Backend Developers
- Frontend Developers

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10. Personalized Roadmap Generator

The roadmap engine identifies gaps between a candidate's current skill level and their desired role.

Workflow

```
Current Skills
        │
        ▼
Gap Analysis
        │
        ▼
Roadmap Builder

        │
        ▼
Learning Plan
```

Example:

Current Skill: Deep

Learning = 35% Target:

Deep Learning = 80%

Generated Roadmap:

Week 1: Neural Networks

Week 2: Backpropagation

Week 3: CNNs

Week 4: Transfer Learning

Database Design

MongoDB is used because candidate profiles naturally fit a document-based structure.

Collections:

- Users
- Resumes
- Assessments
- Candidate Scores
- Recommendations
- Roadmaps

This allows flexible storage of nested candidate information without requiring highly normalized schemas.

Future Scope

Future versions of ResuMate may include:

- Embedding-Based Matching
- Agentic Career Assistants
- Local LLM Integration
- Continuous Learning Pipelines
- Real-Time Interview Evaluation
- Multi-Modal Candidate Assessment

Conclusion

ResuMate is designed as a complete AI-powered career intelligence ecosystem. By combining OCR, NLP, Machine Learning, Recommendation Systems, Predictive Analytics, and Generative AI, the platform aims to transform resumes into actionable career insights while helping candidates validate, improve, and grow their professional skills.

Current Status

Implemented:

- OCR
- Resume Parsing
- NLP Extraction

In Progress:

- Candidate Profile Generation
- Assessment Engine

Planned:

- Job Prediction
- Salary Prediction
- User Clustering
- Roadmap Generation